2 The Earth has a mass M of 5.97×10^{24} kg. A satellite of mass m orbits the Earth in a circular geostationary orbit at a distance d above the Earth's surface.

(a) Show that the radius of the Earth, R, is approximately 6370 km.

$$g = \frac{GM}{R^2}$$

$$\Rightarrow R = \sqrt{\frac{GM}{g}} = \sqrt{\frac{\left(6.67 \times 10^{-11}\right)\left(5.97 \times 10^{24}\right)}{9.81}} \quad \text{[B1]}$$

$$= 6.37 \times 10^{6} \text{ m} = 6370 \text{ km} \quad \text{[A0]}$$

[1]

(b) By considering the orbit of the geostationary satellite, show that the distance d above the Earth's suface is 35900 km.

gravitational force acting on satellite by the Earth
provides the centripetal force on satellite:

$$\Rightarrow \frac{GMm}{r^2} = \frac{mv^2}{r} = \frac{m(r\omega)^2}{r} = \frac{m\left(r\frac{2\pi}{T}\right)^2}{r} \quad \text{[B1]}$$

$$\Rightarrow T^2 = \frac{4\pi^2 r^3}{GM} \Rightarrow r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

$$\Rightarrow r = \sqrt[3]{\frac{\left(6.67 \times 10^{-11}\right)\left(5.97 \times 10^{24}\right)(24 \times 60 \times 60)^2}{4\pi^2}} = 42226.9 \text{ km} \quad \text{[B1]}$$

$d = 42226.9 - 6370 = 35856.9 \text{ km} = 35900 \text{ km (to 3 s.f.)}$ [A0]

[2]

(c) For the satellite of mass m that is in a circular orbit of radius r around the Earth of mass M, show that its total mechanical energy E can be expressed as $E = -\frac{GMm}{2r}$.

gravitational force provides the centripetal force:

$$\Rightarrow \frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow E_k = \frac{1}{2}mv^2 = \frac{GMm}{2r} \quad \text{[B1]}$$

$$E = E_k + E_p$$

$$E = \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right) \quad \text{[B1]}$$

$$E = -\frac{GMm}{2r} \quad \text{[A0]}$$

[2]

(e) The satellite has a mass of 490 kg and carries 360 kg of lithium battery pack with energy density of 324 Wh kg^{-1}, of which 65% of its energy is converted to useful energy for orbital manoeuvres.

(i) Show that the amount of useful energy is 2.73×10^8 J.

useful energy = $360 \times 324 \times 60 \times 60 \times 0.65 = 2.73 \times 10^8$ J [A1]

[1]

(ii) Calculate the maximum orbital radius r_{max} the satellite can reach from its geostationary orbit.

$$\Delta E = E_f - E_i = -\frac{GMm}{2}\left(\frac{1}{r_{max}} - \frac{1}{r_i}\right)$$

$$2.73\times10^8 = -\frac{(6.67\times10^{-11})(5.97\times10^{24})(850)}{2}\left(\frac{1}{r_{max}} - \frac{1}{4.22\times10^7}\right) \quad \text{[M1]}$$

$$\frac{1}{r_{max}} - \frac{1}{4.22\times10^7} = -1.613\times10^{-9}$$

$r_{max} = 45300$ km [A1]

r_{max} = ... m [2]

[Total: 8]